SCRO inductors

$$L = \mu_0 \frac{K_1 n^2 d_{\text{avg}}}{1 + K_2 \rho}$$

$$d_{\text{avg}} = \frac{d_{\text{out}} + d_{\text{in}}}{2}$$

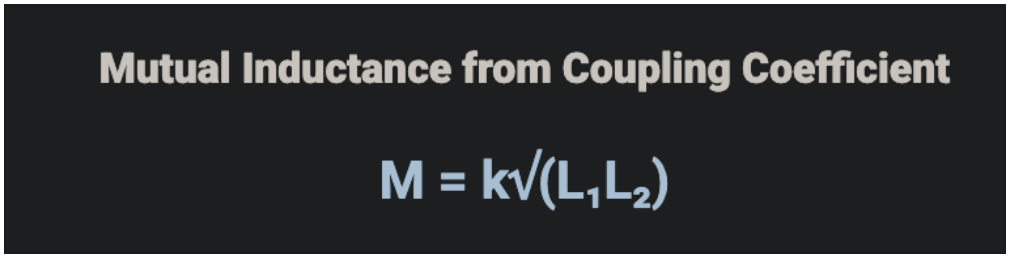
$$\rho = \frac{d_{\text{out}} - d_{\text{in}}}{d_{\text{out}} + d_{\text{in}}}$$

TABLE I
COEFFICIENTS FOR MODIFIED WHEELER EXPRESSION

Layout	K_1	K_2
Square	2.34	2.75
Hexagonal	2.33	3.82
Octagonal	2.25	3.55

-5 micron line
-5 micron spacing between line
-400 micon surface area Max
-mututal inductance matters

Way to measure:
Measure phase shift between two shorted points = LO
Mesure phase shift of circuit = LT
L(Inductor) = LO-LT



LTotal = L1 + L2 + 2M
M=ksqrt(L1*L2)
0<k<1
Match the Resistor: Pick a resistor value close to the expected inductive reactance (\(X_L = 2\pi fL\)) at your test frequency to target a clean **\(45^{\circ }\) phase shift**, where measurement accuracy is highest.